WILSON

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Hands-On Technology Executive — Enterprise Platforms & Production Agentic AI | Start-up Builder · Regulated-Bank Rigour

PROFESSIONAL SUMMARY

Hands-on technology leader with 18+ years of building platforms and leading the teams behind them, across consumer internet, start-ups and a Tier-1 bank. Began as an engineer and technical lead on Renren's e-commerce platform, Sohu's news CMS and Baidu's Nuomi merchant system, then spent 2016–2022 in head-of-technology and architect roles at start-ups and scale-ups: heading technology for a smart-parking IoT platform through its city pilots, co-founding an audio-learning platform, leading VV Technology's technology organisation and building its platforms from scratch, and architecting a 0-to-1 multi-tenant SaaS platform on AWS.

Since 2022, a Vice President in UOB's Group Technology & Operations, leading 25 onshore and offshore engineers on mission-critical payment and lending platforms in a MAS-regulated bank while staying close to the code — reviewing daily, designing solutions and writing core modules. In that role, designed and built a production AI code-review service on headless Claude Code agents and an AI agent for document validation.

Brings start-up speed together with the security, privacy, access control, auditability and reliability that enterprise customers expect. Used to explaining technical choices in business terms to senior management and steering committees, and to taking part in vendor-selection and technology-investment decisions.

CORE COMPETENCIES

Technology Leadership	Technology strategy and roadmap, leading technology organisations at start-ups and scale-ups, leading through squad leads and engineering managers, engineering standards and delivery KPIs, onshore/offshore teams
Applied & Agentic AI	Production LLM agents (headless Claude Code), agentic workflow orchestration, prompt A/B evaluation, AgentOps/LLMOps practice, human-in-the-loop governance aligned with MAS FEAT
Architecture & Engineering	Java, Spring Boot, Spring Cloud, Apache Camel, microservices, event-driven integration, API design, high-concurrency systems, JVM tuning, multi-tenant SaaS
Cloud & Data	AWS (EC2/ECS, API Gateway, Lambda, RDS, DynamoDB, S3, IAM, CloudWatch), Kubernetes, Docker, Kafka, Redis, Elasticsearch, Spark, Hive, MySQL, SQL
Security & Compliance	API and database encryption, credential management, role-based permission design, secure SDLC, DPIA, data governance, MAS TRM (Singapore technology-risk rules)
Reliability & Operations	DR and failover drills, SLIs/SLOs, Splunk monitoring, P1–P3 incident response and RCA, CI/CD quality gates (Jenkins, SonarQube)
Commercial & Stakeholders	Vendor selection and governance, technology-investment decisions, SOW/licence negotiation, technology budget, technical bids and city pilots, executive communication

SELECTED ACHIEVEMENTS

- **Production agentic AI:** Designed and built an AI code-review service in a MAS-regulated bank — a Spring Boot service orchestrates headless Claude Code agents on each pull request, reviewing security, performance, correctness and maintainability alongside SonarQube. Live since May 2026 after formal architecture and risk approval; the AI advises but never approves or merges, and senior engineers' review time fell by an estimated ~80%.
- **AI agent for document validation:** Personally built an AI agent that generates validation prompts to check schema alignment and data-relationship consistency between eMemo and quotation documents, removing a recurring manual reconciliation task.
- **Hands-on technical leadership:** Review code daily across the payment platform's ~50 code modules (deep review of ~20 core modules), co-design solutions with engineers and write core modules for significant changes — including a payment hub's transaction initiation, corporate- and user-role permission matrix and downstream status processing.
- **Scale-up technology leadership:** At VV Technology, led a 140-person technology organisation (total headcount) and built its microservices platform (Spring Cloud, Kubernetes, Redis, Elasticsearch, AWS) and data-analysis platform from scratch, owning technology selection and the technology budget.
- **0-to-1 multi-tenant SaaS:** Architected and delivered, from 0 to 1, a multi-tenant SaaS platform for online-game businesses on AWS — KYC, bonus incentives, VIP tiers, analytics and multi-channel payments — on Spring Cloud and Kubernetes.
- **Enterprise-grade trust:** Designed security controls — API and database encryption of sensitive data, credential management and role-based permissions — for payment, lending and customer-data platforms under secure SDLC, DPIA and MAS TRM; personally led DR and failover drills, sustaining 99.9%+ availability under peak load.
- **Leadership and commercial authority:** Lead two onshore/offshore platform teams through squad leads and engineering managers, with 1:1s, mentoring and performance management; hold decision or joint-decision authority on technology investment, vendor selection and SOW/licence negotiation, and chair governance of four IT service vendors.

PROFESSIONAL EXPERIENCE

United Overseas Bank (UOB) | Singapore — Oct 2022 – Present

Vice President (VP2) / Technical Delivery Manager — Real-Time & Payment Data Services, Group Technology & Operations

Technology leader for two mission-critical platform lines in a MAS-regulated Tier-1 bank — cross-border payments (SWIFT ISO 20022 messaging) and retail unsecured lending — serving millions of accounts.

Applied & Agentic AI

- Took the AI code-review service into production: briefed senior management on the case and the governance model, and improved review quality through prompt A/B testing and AgentOps/LLMOps practices.
- Contributed to a production RAG knowledge assistant used across departments (2025), uploading internal intranet documentation into its vector database, with access controls and data masking in place.
- Coach engineering teams on AI-assisted engineering practices and tooling.

Architecture & Hands-On Engineering

- Led end-to-end design of a payment-initiation hub on Java, Spring Boot and Apache Camel — API contracts, orchestration, validation, enrichment and status tracking — integrating digital and branch channels, reference data, and vendor and third-party banking systems.
- Reduced end-to-end transaction latency under high load through thread-pool and JVM tuning and removal of synchronous blocking.
- Led the 10-engineer backend team of a regional customer-360 platform in production since 2022 — evaluated Cloudera as the processing framework, designed the customer data models and personally built the platform's data-analysis module and Spring Batch scheduled jobs.

Delivery, Reliability & Governance

- Owned delivery of the ISO 20022 (SWIFT MT-to-MX) migration of cross-border payments with a 15-engineer team: a phased, Blue/Green-style coexistence took MX-rich messaging live in May and August 2025, ahead of SWIFT's November 2025 deadline, and retired the coexistence mode in August 2026 with no service interruption; upgrades ship every 2–3 months.
- Coordinate P1–P3 production incidents end to end — escalation, stakeholder communication, RCA and post-incident reports; with no separate SRE team, the platform teams own production support, SLIs/SLOs and Splunk-based monitoring.
- Established delivery KPIs — deployment frequency, MTTR, incident trend and change-failure rate — with CI/CD quality gates (99% SonarQube coverage gate, automated regression and penetration testing) and SIT/UAT sign-off for every release; contributed to bank-wide SDLC, coding and code-review standards.
- Report status, risks and decisions to senior management and steering committees, and work closely with teams in India, Thailand and Indonesia across time zones.

Golden Bridge Drive (SG) Pte. Ltd. | Singapore — Nov 2020 – Jul 2022

Project Manager & Architect

- Led a 40-person project team (total headcount), introduced Agile/Scrum from scratch and CI on Jenkins and Amazon ECR, and owned delivery budget, capacity and launch readiness for the multi-tenant SaaS platform.
- Designed hierarchical KYC identity verification and rule models for authentication levels and for bonus eligibility, triggers and distribution.
- Architected the AWS deployment (API Gateway, EC2/ECS, RDS, DynamoDB, SQS/SNS, CloudWatch) and applied DPIA, API and database encryption and credential management; led data migration.

VV Technology Pte. Ltd. | Singapore — Apr 2019 – Jun 2020

Head of Technical Centre

VV-Life: a local-life services platform connecting offline merchants — food, fresh produce, travel and everyday services — through data.

- Took part in product-strategy planning and delivered the first VV-Life release in 2019, including the payment subsystem's transaction, management and reconciliation modules.
- Designed for performance with Spring Cloud microservices, Kafka-based asynchronous processing and caching that kept load off the database, bringing service response times to the millisecond level.
- Led the big-data team on Flume, Hive, MapReduce, Spark Streaming and Spark SQL, providing data analysis that supported business decisions.
- Designed the middle-platform architecture (Spring Boot, MySQL cluster, Redis, RocketMQ, AWS) for a B2B online-office product serving Southeast Asia, governing solution design, cost evaluation, staffing and acceptance.
- Introduced CI/CD, code standards and containerised deployment with rollback on Kubernetes; defined production-incident priority levels and tiered emergency-response plans.

Hengshunyuan Information Technology Co., Ltd. | China — Oct 2017 – Sep 2019

Co-founder

- Owned product, architecture and technical delivery of an audio-learning platform for workplace and early-childhood content, from market research to production, reaching a peak of ~100k daily active users.
- Brought ~1,500 licensed audio titles and albums onto the platform at launch, on a Spring Cloud microservices stack with a MySQL cluster, Redis, Kafka and ELK.

AI Park | China — Apr 2016 – Sep 2017

VP of Technical Department

- Led architecture, delivery, budget and the technical team for a smart-parking IoT platform connecting car parks to the cloud for automated management, online payment and recommendations; introduced a SaaS model for business and government customers.
- Led technical bids that won city parking projects in Shijiazhuang, Handan and Beijing, and launched pilots in Shijiazhuang (2016), Handan (2017) and Beijing Chaoyang (2017).
- Added an AI processing step to licence-plate and portrait recognition to improve recognition speed and accuracy.
- Brought Spring Cloud microservices into the architecture, with Kafka for asynchronous flows and caching to shield the database, achieving millisecond-level service processing; led the big-data team on HBase, MongoDB and Spark Streaming.

EARLIER EXPERIENCE

- **Baidu — Technical Specialist, Nuomi group-buying e-commerce platform (May 2014 – Dec 2014, China):** As project leader, re-architected the merchant-side order system on Dubbo microservices and delivered two months ahead of plan, leading the web front-end, CMS development and QA teams.
- **Sohu.com — Technical Leader / Project Leader, Sohu News content platform (Mar 2012 – May 2014, China):** Led the re-architecture of the fifth-generation CMS (SSH to Spring MVC; memcached cluster to Redis cluster) and delivered its OpenAPI messaging platform.
- **Renren Network — Senior R&D Engineer (Jul 2008 – Mar 2012, China):** Developed Renren's shopping website, Renren Mall and CRM.

EDUCATION, CERTIFICATIONS & LANGUAGES

- Master's degree in Computer Science and Technology — Liaoning University (2005 – 2008)
- Bachelor's degree in Applied Mathematics — Shenyang University of Technology (2000 – 2004)
- Software Designer (Intermediate) — Qualification Certificate of Computer and Software Technology Proficiency (2006) · PMP — in progress · Site Reliability Engineering (SRE) — formal training
- Languages: English and Mandarin (professional working proficiency in both)